

HARRIET ALEXANDER FIELD, CO, USA

WMO: 720532

Lat: 38.533N

Lon: 106.050W

Elev: 2294

StdP: 76.63

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 00159

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-16.3	-13.2	-23.9	0.6	-3.9	-22.1	0.7	-3.0	13.4	7.8	12.2	4.9	2.3	130	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.1	31.7	12.1	30.5	12.0	29.1	11.9	15.2	22.9	14.7	22.7	14.1	22.5	4.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.3	12.7	16.8	12.4	11.9	16.7	11.4	11.1	16.9	51.3	22.6	49.4	22.5	47.6	22.5	17.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.7	10.4	9.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-0.2	0.4	5.6	8.1	12.2	19.4	21.3	19.9	16.9	10.1	4.1	-0.4
	DBStd	8.80	4.66	5.54	4.35	4.22	3.90	2.78	1.81	1.91	2.94	4.52	4.85	5.43
	HDD10.0	1392	316	271	146	83	23	0	0	0	1	52	179	322
	HDD18.3	3320	574	503	396	308	189	21	2	8	57	255	426	580
	CDD10.0	1335	0	1	8	24	93	283	351	307	208	55	4	0
	CDD18.3	222	0	0	0	0	1	53	94	57	15	1	0	0
	CDH23.3	3106	0	0	0	4	72	941	1051	702	322	14	0	0
	CDH26.7	994	0	0	0	0	4	341	390	198	62	0	0	0
(14) Wind	WSAvg	3.8	4.1	4.7	4.3	4.1	3.7	3.8	3.1	3.0	3.2	3.7	4.1	4.2
(15) Precipitation	PrecAvg	293	14	14	22	31	27	19	39	47	27	22	16	14
	PrecMax	368	24	37	47	95	77	37	79	104	50	50	38	53
	PrecMin	175	4	4	4	5	2	4	8	14	9	5	4	4
	PrecStd	57	6	9	13	24	21	10	20	21	12	13	9	11
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	15.2	21.3	23.7	27.3	33.7	32.9	31.6	30.5	25.1	18.8	14.0
		MCWB	3.2	4.5	6.4	7.6	8.8	12.1	12.6	11.7	12.0	9.5	5.9	3.0
	2%	DB	11.2	12.9	19.1	21.3	25.3	31.8	31.7	30.3	28.4	22.8	16.7	11.6
		MCWB	1.9	3.0	5.5	6.6	8.3	11.7	12.6	12.0	11.0	8.1	5.0	2.2
	5%	DB	9.0	10.7	16.9	19.2	23.8	30.4	30.5	28.9	26.9	21.1	14.8	9.7
		MCWB	1.2	2.0	4.5	6.0	8.0	11.2	12.6	12.3	10.7	7.5	4.1	1.6
	10%	DB	7.3	8.7	14.0	17.6	21.9	28.7	29.0	27.5	25.2	19.0	12.5	7.7
		MCWB	0.2	1.4	3.2	5.3	7.6	10.7	12.8	12.4	10.3	7.1	3.0	1.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.7	6.8	8.6	11.5	15.1	16.2	15.7	14.7	11.4	6.9	4.2
		MCDB	11.2	13.8	19.3	19.0	19.0	25.1	23.1	22.8	21.1	19.6	15.5	10.8
	2%	WB	2.8	3.6	5.8	7.6	10.0	13.9	15.5	14.9	13.6	10.1	5.6	3.1
		MCDB	9.0	11.2	17.2	19.0	19.4	23.7	22.9	22.5	20.3	19.0	14.5	9.2
	5%	WB	1.7	2.7	4.9	6.8	9.1	13.0	15.0	14.3	12.8	8.8	4.6	2.3
		MCDB	7.8	9.6	15.2	17.4	19.8	24.1	22.8	22.5	20.7	17.6	13.6	8.7
	10%	WB	0.7	1.6	3.8	5.8	8.4	12.3	14.5	13.8	12.1	7.8	3.5	1.2
		MCDB	6.1	8.2	12.7	15.8	19.4	24.0	22.8	22.3	20.8	17.0	11.8	7.5
(35) Mean Daily Temperature Range	MDBR		12.3	11.7	14.1	14.8	15.0	17.5	16.1	15.9	15.7	14.8	13.0	11.6
	5% DB	MCDBR	14.4	13.0	17.2	17.6	18.0	19.0	18.2	17.8	18.3	17.3	15.5	13.4
		MCWBR	8.1	6.6	7.5	7.3	6.6	6.3	5.1	5.3	6.1	7.1	7.3	7.1
	5% WB	MCDBR	12.6	11.9	15.8	16.5	16.6	16.8	15.2	14.9	14.2	14.9	14.6	11.4
		MCWBR	7.3	6.3	7.3	7.1	6.3	5.7	4.7	4.8	4.8	6.2	7.1	6.4
(40) Clear-Sky Solar Irradiance	taub		0.228	0.236	0.265	0.286	0.301	0.307	0.348	0.333	0.288	0.266	0.241	0.232
	taud		2.524	2.502	2.484	2.491	2.517	2.514	2.451	2.466	2.603	2.621	2.605	2.537
	Ebn at Noon		985	1014	1005	994	977	966	923	933	967	966	963	955
	Edh at Noon		73	87	99	105	105	105	111	107	87	77	68	68
(44) All-Sky Solar Radiation	RadAvg		2.81	3.78	5.08	5.99	6.52	7.29	6.40	5.84	5.32	4.12	3.04	2.44
	RadStd		0.14	0.21	0.38	0.37	0.49	0.43	0.34	0.27	0.31	0.27	0.17	0.14

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+73	+20

Nomenclature: See separate page